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Mask Optimization for Image Inpainting Using No-Reference Image Quality Assessment

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ABSTRACT Image inpainting is a technique designed to remove unwanted regions from images and restore them. This technique is expected to be applied in various applications, including image editing, virtual reality (VR), mixed reality (MR), and augmented reality (AR). Typically, the inpainting process is based on missing regions predefined by user-applied masks. However, the specified areas may not always be ideal for inpainting, and the quality of the inpainting results varies depending on the annotated masked region. Therefore, this paper addresses the task of **generating masks that improve inpainting results**. To this end, we proposed a method that utilized No-Reference Image Quality Assessment (NR-IQA), which can score image quality without a reference image, to generate masked regions that maximize inpainting quality.

INDEX TERMS Image inpainting, mask optimization.

I. INTRODUCTION

Image inpainting, also known as image completion, is a technique that removes specified regions in images and restores them to a natural appearance. This technique holds great potential for various applications, including general image editing, VR, MR, and AR. To date, many effective learning-based inpainting methods, including those using GANs and Diffusion Models [1], have been proposed [2], [3]. These approaches have improved the quality of inpainted images as well as significantly expanded the applicability of inpainting.

One key aspect of inpainting is identifying the region to be restored (hereafter, the masked region). In many inpainting methods, users manually annotate the masked region, which is **not always optimized for the inpainting process**. For instance, the user-provided mask may not precisely cover the required region for restoration, or it may encompass an unnecessarily large area. Thus, user-provided masks can lead to subjective inconsistencies results from regions remaining unprocessed or poor restoration quality due to insufficient contextual information outside the mask.

Therefore, this paper challenges to **optimize masked regions to improve inpainting quality**, as shown in Fig. 1. Specifically, we proposed a learning-based masked region optimization method that modifies a given masked region to produce improved inpainting results. Previous work proposed a similar task based on super-pixel level image quality assessment using ranking learning [4]. Other methods have been proposed to improve inpainting quality by connecting a segmentation network to the inpainting network and training them end-to-end for mask optimization [5].

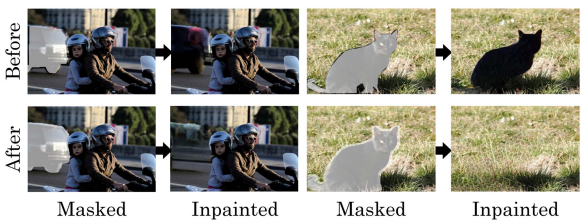

FIGURE 1. Mask Optimization: The optimization of the masked region improves the quality of inpainting.

However, the former method has the limitation that mask optimization can only be performed at the super-pixel level, while the latter method is not suitable for optimizing masks for fine shapes. Additionally, this latter method does not consider the human perception of inpainting quality, i.e., how natural the image looks.

In contrast, this paper proposed a deep-learning-based approach reflecting human perception to achieve pixel-level masked region optimization. Specifically, we developed a framework that generates an optimal masked region to maximize inpainting quality given only an original image and an initial masked region. Since inpainting is a *task without absolute ground truth*, for which the goal is to create images that are natural to human perception, we assessed the inpainting quality using No-Reference Image Quality Assessment (NR-IQA), a quality assessment method that does not require reference (ground truth) images.

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This paper makes the following contributions: (1) We tackle the task of "masked region optimization," which deforms a given initial masked region to achieve better inpainting results. Our method is the first to enable pixel-level optimization while considering human perception. (2) To this end, our framework introduces a deep-learning-based approach for pixel-level optimization and NR-IQA to assess human perception-aware inpainted image quality. (3) This paper demonstrates the effectiveness of our method by comparing the optimized mask generated by our approach with the initial and dilated masks, thereby confirming improvements in the restoration region.

II. RELATED WORK

A. IMAGE INPAINTING

Image inpainting is a technique that removes specified areas from an image and restores them. Recently, numerous effective image inpainting methods have been proposed, particularly those that leverage image generation models. Image inpainting has been extensively studied since before the advent of deep learning. The concept of digital inpainting was popularized by Bertalmio et al. [6], while earlier approaches, such as the level lines-based disocclusion method proposed by Masnou and Morel [7], represented important milestones in terms of model-based techniques. For a comprehensive overview of both traditional and recent inpainting methods, we refer readers to the review by Guillemot et al. [8]. EdgeConnect [9] is an inpainting method inspired by the way artists work: lines first, color next. EdgeConnect is a two-stage adversarial model consisting of an edge generator followed by an image completion network. The edge generator predicts the edges of the missing regions in an image, and the image completion network fills in these missing regions using the predicted edges as prior. Also, LaMa [2] is an inpainting method, that employs Fast Fourier Convolution (FFC) with a global receptive field. It stands out for its use of the image's global context. LaMa incorporates a perceptual loss based on a semantic segmentation network with a broad receptive field. It demonstrates excellent performance with relatively few parameters. However, a common challenge of these existing deep-learning-based inpainting methods is their exclusive focus on restoration techniques, with little to no effort directed at optimizing masked regions. To address this gap, this paper focuses on optimizing masked regions for inpainting. To restore images with our proposed method, we utilized EdgeConnect and LaMa as the network backbone.

B. THE EFFECT OF THE MASKED REGION

Recently, methods such as Inpaint Anything [10] and SAMRefiner [11] have explored mask refinement pipelines using the Segment Anything Model (SAM) [12]. These approaches focus on leveraging high-quality semantic segmentation priors to improve user-specified or automatically generated masks. In the context of inpainting, it is well known that the selection and accuracy of the masked region significantly influence the quality of the final result. This is because the mask defines which parts of the image are to be edited, and the model relies on this information to perform the completion. Consequently, an inappropriately masked region leads to lower-quality inpainting results.

The Importance of Mask in Inpainting: Previous works have pointed out that the restoration accuracy of inpainting methods can vary depending on how the masked regions are provided. In EdgeConnect, two types of image masks—regular and irregular—are employed, and their results are compared. Regular masks are applied as square masks at random positions within the image, while irregular masks are derived from Liu et al.'s research [13] and are classified in 10% increments based on their size relative to the entire image. As revealed in this paper, as the size of the masked region increases, inpainting performance deteriorates both quantitatively and qualitatively. Additionally, the authors reported a tendency for performance to decline when using irregular masks, which are closer to real-world scenarios than regular masks. Importantly, the authors of LaMa demonstrated that the method of mask generation has a significant impact on the final performance of the LaMa model. It adopts an *aggressive large mask generation strategy*, which uniformly samples from polygonal chains dilated by a high random width (wide masks) and rectangles of arbitrary aspect ratios (box masks). This paper, it compares the results of training inpainting methods using both large masks and narrow masks. The results demonstrated that training with large masks achieved the highest level of accuracy. Moreover, high-quality restoration was achieved regardless of whether large masks or narrow masks were used during prediction. Thus, the shape of the masked region had a significant impact on the performance of inpainting.

Relationship to Blind Image Inpainting: Another line of work relates to blind image inpainting, which refers to the task of inferring missing or damaged regions directly from the image content without any externally provided mask. Notable examples include deep-learning-based approaches such as those developed by Cai et al. [14], Liu et al. [15], Wang et al. [16], and InViT [17], which leveraged convolutional or transformer architectures to localize and restore unknown regions simultaneously. These methods treat the mask as either entirely unknown or weakly defined and estimate it in an unsupervised or weakly supervised fashion. In contrast, our proposed method assumed the availability of an initial mask—such as one provided by a user or a preprocessing step—and aimed to deform it to optimize the quality of the inpainting. This makes our approach more suitable for practical scenarios, in which users provide coarse masks that require refinement. While both paradigms involve a form of mask estimation, blind image inpainting addresses localization from scratch, whereas our method refines given regions based on perceptual quality assessment.

Optimization of Masked Regions for Inpainting: The performance of image inpainting is significantly influenced by the masked region. However, little attention has been given to the optimization of the masked region itself. Inpaint Anything simply applies uniform dilation to the masks without any optimization step. Existing methods that address this task include the work by Isogawa et al. [4] and Himosato et al. [5]. In particular, Isogawa et al. optimized the masked region based on subjective quality estimation at the super-pixel level using a ranking-based learning approach. However, this method has the limitation of only optimizing the mask at the super-pixel level. On the other hand, Shimosato et al. proposed an inpainting network that utilized object masks extracted via segmentation, connecting the segmentation network to the inpainting network for end-to-end training. Additionally, they introduced a mask expansion loss in their framework to resolve the trade-off caused by varying mask sizes, enabling optimization of the masked region. However, this approach does not incorporate image quality evaluation into the training process, resulting in an optimization that does not consider human perception. In contrast, we proposed an optimization method for the masked region that leverages NR-IQA to evaluate the quality of the inpainting results and incorporate this evaluation into the training process. This approach aims to guide the generation of more natural inpainting results that align with human perception.

The Relationship to Sparse Mask Design for Compression and Sampling: Another area of research that uses the term "mask optimization" focuses on identifying sparse data selection patterns for applications such as image compression, adaptive sampling, and efficient reconstruction. In such settings, the entire original image is typically known, and the goal is to select the most informative subset of pixels or regions that preserve structural or tonal fidelity during encoding or diffusion-based recovery. In contrast, our approach addresses the problem of deforming an externally provided incomplete mask to improve the perceptual quality of the restored content in image inpainting, where parts of the image are genuinely unknown. Thus, while the terminology overlaps, the objective and data assumptions are fundamentally different. We refer readers to Peter et al. [18] for a comprehensive overview of sparse mask optimization techniques.

C. NO-REFERENCE IMAGE QUALITY ASSESSMENT (NR-IQA)

To achieve high-quality inpainting results, it is crucial to assess the quality of the inpainting. However, inpainting is a task that lacks ground truth images. Consequently, evaluating restoration quality depends on subjective assessment, making it difficult to use objective measures. Therefore, this paper adopts NR-IQA as the evaluation metric for inpainting quality. NR-IQA is a quality assessment method designed to evaluate image quality in situations where no reference images (i.e., ground truth, original, or high-quality images) are available. Most NR-IQA methods aim to estimate image quality while considering human subjective perception.

Q-Align [19] deviates from the conventional numeric-based evaluation approaches by employing discrete text-defined levels (e.g., 'excellent', 'fair', or 'bad') to train Large Multi-modality Models (LMMs). This method has achieved state-of-the-art performance in Image Quality Assessment (IQA), Image Aesthetic Assessment (IAA), and Video Quality Assessment (VQA). Additionally, Neural Image Assessment (NIMA) [20], developed by Google, is a CNN-based no-reference IQA model that is trained to evaluate both the technical quality and artistic beauty of images. NIMA employs a dataset in which each image is annotated with a score ranging from 1 to 10 based on its aesthetic appeal, from which the CNN learns from during the training process. Other NR-IQA methods include HyperIQA [21], which employs a self-adaptive hypernetwork architecture to perform blind image quality assessment in the wild, and UNIQUE [22], which is trained to handle both synthetic and realistic distortions. Another approach is the patch-based Multi-scale Image Quality Transformer (MUSIQ) [23], which effectively predicts image quality at native resolution, thereby avoiding the fixed input size constraints of CNNs. CLIP-IQA [24] explores the rich visual language priors encapsulated in Contrastive Language-Image Pre-training (CLIP) [25] models to assess both the quality perception (i.e., how an image looks) and abstract perception (i.e., how an image "feels") of images in a zero-shot manner. The Multi-dimension Attention Network for No-reference Image Quality Assessment (MANIQA) [26], based on GANs, enhances robustness to distortions, while the Language-Image Quality Evaluator (LIQE) [27] not only performs NR-IQA but also includes scene classification and distortion-type identification, such as "blur." Unlike other bottom-up methods, TOPIQ [28] adopts a top-down approach that leverages high-level semantic information to emphasize low-level distortions while maintaining computational efficiency. Additionally, RAPIQUE [29] has been proposed as a No-Reference Video Quality Assessment (NR-VQA) model, effectively capturing both spatial and temporal features to assess the quality of user-generated content (UGC) videos.

In this study, we employed Q-Align, which is more accurate and faster in execution compared to other methods, for evaluating the quality of inpainting results in the Inpainting Module described in Section III-B. Additionally, in the Mask Dilation Module, as described in Section III-D, where numerous image quality assessments must be performed on small image regions to generate dilated masks, we employed the lightweight NIMA model to ensure computational efficiency.

III. PROPOSED METHOD

A. PROPOSED FRAMEWORK

Fig. 2 provides an overview of our proposed framework. It takes an input image I_{input} and an initial mask M_{ini} as inputs and generates a masked region M_{pred} that maximizes the image quality assessment score of the inpainted result. The image quality assessment score of the inpainted result is calculated using an NR-IQA model. Each step of the proposed method is detailed below. First, a four-channel tensor T_{concat} is created by concatenating the input image I_{input} and the initial masked region M_{ini}. Here, M_{ini} is assumed to be a given masked region, and the method for generating M_{ini} is described in detail in Section IV-D. Next, T_{concat} is fed into a segmentation network, which outputs the pixel-wise class probability map P_{pred}. Subsequently, P_{pred} is input into the inpainting module to generate the inpainted image $I_{inpainted}$. We utilized U-Net [30] as the segmentation network for generating masks. The parameters constituting the proposed model are learned according to (i) the inpainting score loss $\mathcal{L}_I$, which evaluates the inpainted image $I_{inpainted}$ using an NR-IQA model, and (ii) the mask shape consistency loss $\mathcal{L}_M$, which assesses P_{pred} to maintain the shape of the mask. To prevent the generation of M_{pred} that significantly deviates from M_{ini}, we introduced (ii). Additionally, the initial mask M_{ini} may not fully cover the objects that need to be restored. To create ground truth masks that properly cover the target objects, we employ the Mask Dilation Module to generate a dilated mask $M_{dilated}$ from M_{ini}. The Inpainting Module is discussed in Section III-B, and the loss functions are explained in Section III-C. Finally, the Mask Dilation Module is detailed in Section III-D.

B. INPAINTING MODULE

This module generates an inpainted image $I_{inpainted}$ given an input image I_{input} and the pixel-wise class probability map P_{pred} output by the segmentation network. Since inpainting networks typically require an image and a binary mask indicating the regions to be restored as input, we convert the probability map P_{pred} into a binary prediction mask M_{pred}. In conventional segmentation tasks, a fixed threshold is applied to each pixel to obtain a binary mask. However, such standard thresholding is not suitable for our approach—which relies on gradient-based optimization—as it is non-differentiable. To address this issue, we employ the Straight-Through Estimator (STE) [31], [32], enabling differentiable thresholding. Specifically, during the forward pass, standard thresholding is applied, while during the backward pass, the gradients are propagated directly, thereby allowing for the generation of a differentiable prediction mask. This approach facilitates end-to-end training and is expected to enhance segmentation performance.

In this study, we generate predicted binary masks based on the $2 \times 256 \times 256$ output tensor from the U-Net used as the segmentation

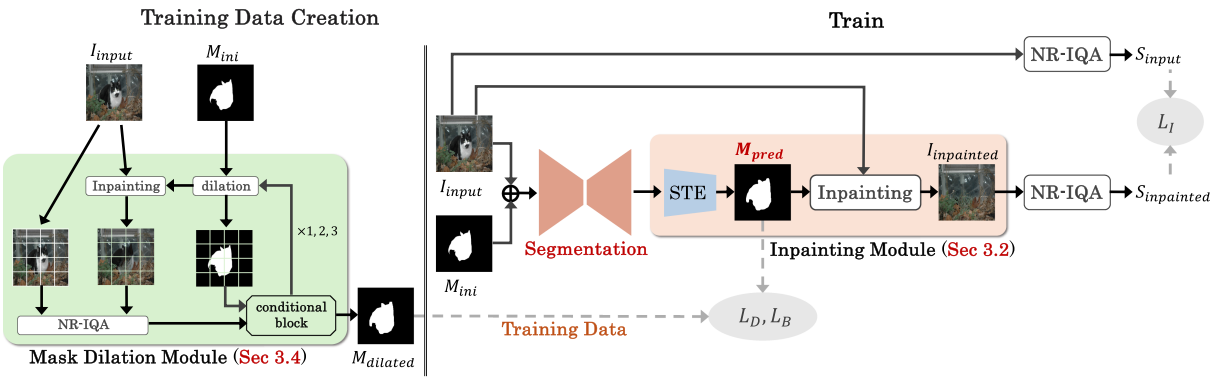

FIGURE 2. Proposed Framework Overview. A masked region M_{pred} is generated to maximize the quality score of the inpainting results using the input image I_{input} and the initial mask M_{ini}.

network. First, the softmax function is applied to the output tensor to obtain the probability distribution for each class. Then, to generate binary masks, we employ STE to convert the logits into binary values. This binary output determines whether each pixel belongs to the target object in the segmentation task. The resulting binary mask is then output as the final segmentation result. Additionally, during training, we employ EdgeConnect and LaMa as inpainting models, and the training is conducted separately for each model.

FIGURE 3. The masks output when training is performed using only inpainting loss. A low-quality mask is generated.

C. LOSS FUNCTION

Inpainting Score Loss: We introduce an inpainting score loss $\mathcal{L}_I$, which utilizes an NR-IQA metric to objectively evaluate the effectiveness of the predicted mask M_{pred} in image inpainting tasks. Specifically, we obtain NR-IQA scores for both the input image I_{input} and the inpainted image $I_{inpainted}$. Then, $\mathcal{L}_I$ takes the differences of these scores. By denoting the quality scores of I_{input} and $I_{inpainted}$ as S_{input} and $S_{inpainted}$, respectively, the loss function $\mathcal{L}_I$ can be expressed as follows (1):

$$\mathcal{L}_I = \begin{cases} S_{input} - S_{inpainted} & S_{input} \geq S_{inpainted} \\ 0 & S_{input} < S_{inpainted} \end{cases} \tag{1}$$

Here, the NR-IQA score indicates higher naturalness with higher values. Equation (1) means that the loss is set to zero when $I_{inpainted}$ is assessed as more natural than I_{input} with a higher score since it indicates that the M_{pred} leads better inpainting result. Otherwise, the loss function incorporates the difference by subtracting the NR-IQA score of I_{input}, denoted as S_{input}, from that of $I_{inpainted}$, denoted as $S_{inpainted}$. Note that both S_{input} and $S_{inpainted}$ are computed on the entire image, including the masked region. The input image I_{input} contains the original (possibly corrupted) pixels in the masked area, and the comparison evaluates the perceptual naturalness of the overall image before and after inpainting. This allows the NR-IQA metric to reflect how successfully the inpainting improves global visual quality. During training, we employ Q-Align, which is capable of evaluating both IQA and IAA. When IAA was used during training, the model tended to minimize the mask value to zero, aiming to make the output image resemble the input image, which was rated as having the highest quality. Therefore, in this study, we incorporate the IQA metric of Q-Align into the loss function as NR-IQA.

Mask Shape Consistency Loss: When training is performed using only the loss function $\mathcal{L}_I$, the predicted mask M_{pred} expands rapidly and takes on a speckled pattern, resulting in an output where M_{pred} lacks any coherent shape, as shown in Fig. 3. To prevent the predicted mask M_{pred} from deviating significantly from the initial mask M_{ini}, we incorporated two additional loss functions commonly used in segmentation tasks: the Dice Loss $\mathcal{L}_D$ and the Boundary Loss $\mathcal{L}_B$ [33]. These losses were employed to maintain consistency between M_{pred} and M_{ini} in terms of shape and structure. The Dice Loss is specifically designed to facilitate effective learning even with imbalanced data, and it is commonly used in segmentation tasks. This loss is computed using the Dice coefficient, which measures the overlap between the ground truth (GT) and the predicted regions. In addition, the Boundary Loss takes into account the distance between boundaries and, when combined with the Dice Loss during training, has been reported to improve both accuracy and learning stability, particularly with highly imbalanced data. A straightforward approach could involve comparing the areas of M_{ini} and M_{pred}, ensuring that the difference between them remains within a certain range. However, using M_{ini} as ground truth data may not be appropriate because it might not fully cover the regions that need to be restored. In our proposed method, we use the dilated mask $M_{dilated}$, which is created by expanding M_{ini}, as the ground truth data. We generated $M_{dilated}$ using two different approaches. The first involves uniformly dilating the mask, while the second utilizes the Mask Dilation Module, as described in Section III-D, to achieve the dilation. The Dice Loss and Boundary Loss are expressed in (2) and (4), respectively. The loss function $\mathcal{L}_M$ is calculated as the sum of $\mathcal{L}_D$ and $\mathcal{L}_B$.

$$\mathcal{L}_D = 1 - \frac{2\Sigma_i p_i g_i}{\Sigma_i (p_i + g_i)} \tag{2}$$

where the sums run over the pixels, of the predicted segmentation pixel $p_i \in P_{pred}$ and the ground truth binary label $g_i \in M_{dilated}$

$$\phi_{M_{dilated}}(p) = \begin{cases} -D_{M_{dilated}}(p), & \text{if } p \in M_{dilated}, \\ D_{M_{dilated}}(p), & \text{otherwise.} \end{cases} \tag{3}$$

where $D_{M_{dilated}}(p)$ is the distance map from the boundary of $M_{dilated}$.

$$\mathcal{L}_B = \int_{\Omega} \phi_{M_{dilated}}(p) P_{pred}(p) dp \tag{4}$$

where p, Ω, and $P_{pred}(p)$ denote a point on the boundary of $M_{dilated}$, a pixel set in the image, and the sigmoid output of the segmentation network at p. $\phi_{M_{dilated}}(p)$ is computed as described in (3).

The overall loss function $\mathcal{L}$ is formulated using the hyper-parameters α, β and γ as follows:

$$\mathcal{L} = \alpha \mathcal{L}_I + \beta \mathcal{L}_D + \gamma \mathcal{L}_B \tag{5}$$

D. MASK DILATION MODULE

To generate training data where the predicted mask region covers the object to be restored, the Mask Dilation Module performs iterative processing (loop processing). This process creates a dilated masked region $M_{dilated}$ from the original image I_{input} and the input masked region M_{ini}. Specifically, the process begins with inpainting using the mask image. Subsequently, the original image, the mask image, and the inpainted image are divided into patches. For patches containing the mask boundary, the quality is evaluated using NR-IQA. If the quality score of the original image is greater than that of the inpainted image for a corresponding patch, the masked region in the corresponding patch is dilated. The loop terminates either when the quality score of the original image is lower than that of the inpainted image across all patches or when the maximum iteration count of three is reached. The flow of this process is illustrated in the Supplementary Material. Note that the NR-IQA evaluation is conducted on visually complete image patches, not on masked or zero-filled images. This ensures valid perceptual quality estimation. The score comparison is used as a signal to determine whether the current masked region is sufficient: if the inpainted patch degrades perceived quality, the masked region is expanded to include more relevant context. This strategy allows the mask to grow adaptively based on restoration performance rather than static heuristics. In this module, when evaluating image quality, NIMA is selected as the NR-IQA metric due to its lightweight nature and suitability for patch-based evaluation. This component requires frequent computation on small image regions, and NIMA provides a good balance between speed and quality for such use cases.

IV. EXPERIMENTAL SETUP

A. BASELINE METHODS

No existing study has optimized the most suitable masked regions for inpainting based on the a given image and mask. Therefore, we compared our method against other baselines, including methods that use the initial masks (M_{ini}), which are assumed to be provided externally and used as input to the model, as well as masks created by uniformly dilating M_{ini}.

B. EVALUATION METRICS

To evalute mask optimization for inpainting, NR-IQA was employed as metric for the inpainted images generated using the original image and the mask. This approach was chosen because directly assessing the generated mask itself is challenging, and the inpainted images lack clear ground truth data. Therefore, NR-IQA, which does not require a reference image, was deemed appropriate for assessing how natural the generated images appeared after inpainting. We used four types of NR-IQA metrics to evaluate our method: Q-Align, TOPIQ, LIQE, and MANIQA. While Q-Align was used during training, TOPIQ, LIQE, and MANIQA were adopted as evaluation methods because they are excellent NR-IQA techniques. These metrics are presented in the listed order, with the earlier ones being more recent than the later ones.

C. IMPLEMENTATION DETAILS

We employed the Adam optimizer with a learning rate of 10^{-4} over five epochs, and a batch size of 16. The loss function weights were set as $\alpha = 1$, $\beta = 0.0095$, and $\gamma = 0.0005$. In the experimental environment, we utilized 30 GiB of RAM, an Intel(R) Xeon(R) w5-3435X CPU, and an NVIDIA RTX 6000 Ada as a GPU. The training of our proposed mask optimization model (excluding the inpainting network) required approximately two hours for five epochs in this environment. For inference, generating an optimized mask M_{pred} from an input image and M_{ini} took 20.5ms per image on average. We employed EdgeConnect and LaMa as the inpainting backbones due to their moderate computational cost, ease of integration, and stable training behavior. We considered more recent diffusion-based or transformer-based inpainting models, but their substantial resource requirements exceeded our experimental constraints. However, the proposed framework was modular and compatible with any inpainting network that accepts masks as input.

D. DATASET

To obtain images and corresponding masks, we utilized the Common Objects in Context (COCO) dataset [34], a widely used multi-purpose dataset for computer vision tasks. This dataset contains a wide variety of objects, ranging from animals such as cats and dogs to vehicles such as cars and buses. When considering the application of masks to cover these objects, animals with features like fur present more challenges in terms of accurately capturing their contours compared to solid objects. The difficulty of applying masks varies based on the characteristics of the contours (i.e., whether the boundaries are clearly defined). To address this, we created two datasets based on the difficulty of the mask application: the "Animal Dataset" and the "Object Dataset." The former dataset utilized images and annotations from the "cat" category of the COCO dataset, while the latter dataset utilized those from the "car" category of the same dataset. From this selection, we extracted images where the area designated for inpainting covered between 2% and 40% of the entire image. In the "Animal Dataset", 3,310 images were used for training and 151 images for testing, while in the "Object Dataset", 4,198 images were used for training and 174 images for testing. In the training dataset, 90% of the images were used as the training set, while the remaining 10% were allocated for validation. The initial mask is intended to represent an externally input restoration area; however, such masks rarely extract the contours cleanly. They may either fail to completely cover the object to be removed or excessively cover surrounding areas. Since the segmentation in the dataset is annotated along the object's contours, it is not appropriate to use it directly as the initial mask. To address this issue, we generated variations of the initial masks by applying random dilation and erosion processes to the annotated masks. Specifically, we randomly selected one of three options—dilation, erosion, or no change—at a ratio of 7:2:1 and applied the corresponding image processing. For dilation, a kernel size was randomly chosen between 1 and 49, and the mask was dilated using this kernel. For erosion, the kernel size was randomly selected between 1 and 9, and the mask was eroded accordingly. The dilation and erosion operations were performed using OpenCV. This approach enabled the creation of more realistic restoration regions

TABLE 1. Quantitative Comparison

Masks used for inpainting	Animal Dataset				Object Dataset			
	Q-Align (↑)	TOPIQ (↑)	LIQE (↑)	MANIQA (↑)	Q-Align (↑)	TOPIQ (↑)	LIQE (↑)	MANIQA (↑)
M_{ini}	2.480	0.6072	3.694	0.4225	2.690	0.6709	4.136	0.4517
Uniformly Dilated M_{ini}	2.413	0.6080	3.580	0.4127	2.701	0.6702	4.154	0.4527
M_{pred} w/ EdgeConnect	2.484	**0.6097**	3.705	**0.4231**	**2.755**	**0.6782**	**4.231**	**0.4589**
M_{pred} w/ LaMa	**2.487**	0.6090	**3.712**	0.4229	2.712	0.6736	4.171	0.4531

The bold values indicate the highest performance scores among the methods compared in each row.

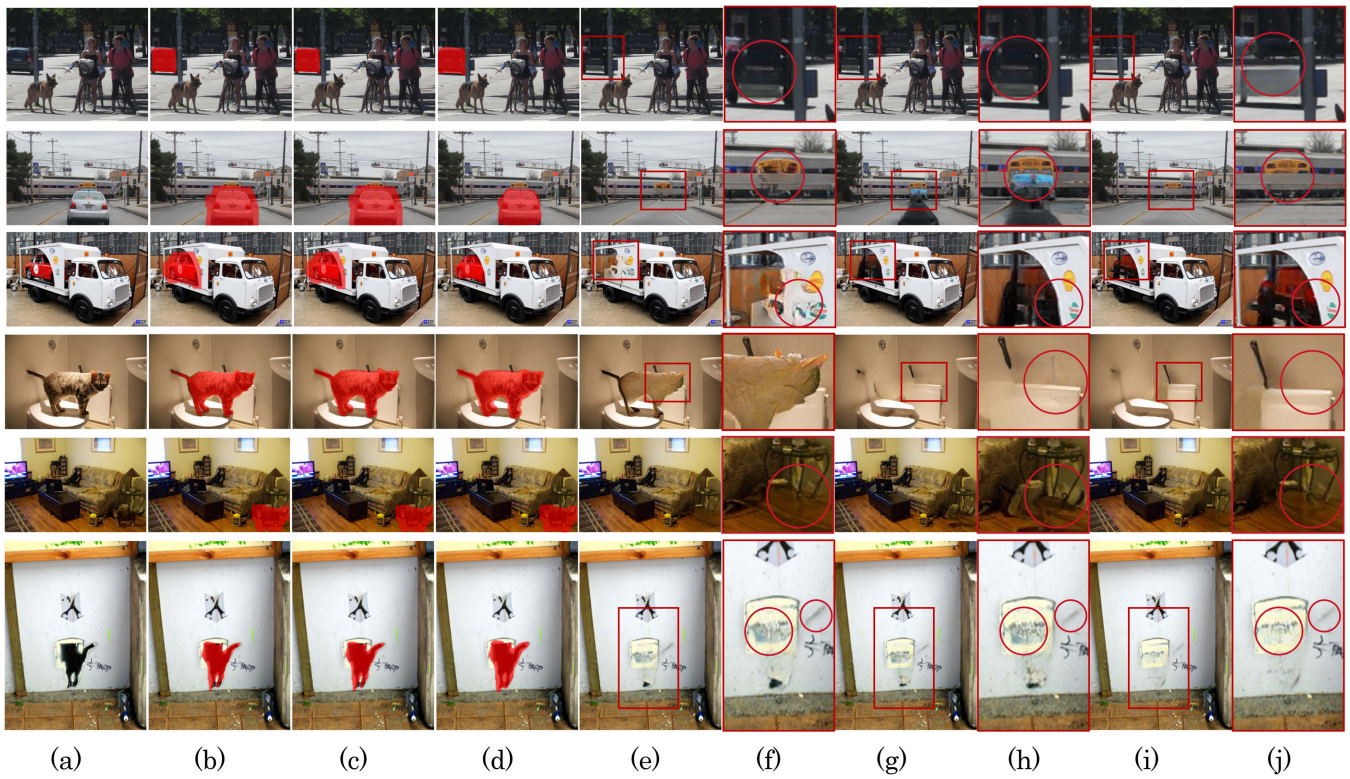

FIGURE 4. Comparison of inference results: (a) I_{input} (Original Image), (b) M_{ini}, (c) Uniformly dilated M_{ini}, (d) M_{pred} (Mask Generated by the Proposed Method), (e) and (f) Inpainting results using M_{ini} and enlarged view, (g) and (h) Inpainting results using uniformly dilated M_{ini} and enlarged view, (i) and (j) Inpainting results using M_{pred} and enlarged view.

that could be provided externally, better reflecting real-world scenarios.

V. EXPERIMENTS AND RESULTS

We conducted four different experiments to investigate our method's efficacy: (1) a comparison against baselines; (2) an experiment to measure the effect of the Masked Dilation Module; (3) a comparison with a method that modified the calculation of the Inpainting Score Loss; and (4) a user study on inpainting results optimized for masks. In all of the experiments, LaMa was used as the inpainting method during prediction.

A. COMPARISON AGAINST BASELINE METHODS

We conducted a comparison between the proposed method and baseline methods using two different datasets to demonstrate the effectiveness of mask optimization for inpainting. It should be noted that **the Mask Dilation Module was not used during training in this experiment. Instead, uniformly dilated M_{ini} were used as the training data.** Table 1 presents the quantitative evaluation results compares the best model of our method with the baseline approaches on both datasets. Specifically, we report performance using different inpainting methods—EdgeConnect and LaMa—during training. The best value is indicated in **bold**, while the second-best value is underlined. As shown in Table 1, using the masks generated by our proposed method (M_{pred}), we achieved higher accuracy across all NR-IQA evaluation metrics compared to the baseline approaches, M_{ini}, and uniformly dilated M_{ini}. It can be concluded that our method successfully optimized the masked regions. Furthermore, the results of the training with both EdgeConnect and LaMa demonstrated higher accuracy compared to the baseline methods, indicating that our approach is applicable regardless of the inpainting method used. Fig. 4 illustrates the qualitative results in the cross-subject setting. The original image I_{input} is displayed in (a), while (b), (c), and (d) show, respectively, the initial mask M_{ini}, the mask created by uniformly dilating M_{ini}, and the mask generated by our proposed

TABLE 2. The Effect of the Mask Dilation Module

Masks used for inpainting	Animal Dataset				Object Dataset			
	Q-Align (↑)	TOPIQ (↑)	LIQE (↑)	MANIQA (↑)	Q-Align (↑)	TOPIQ (↑)	LIQE (↑)	MANIQA (↑)
Uniformly Dilated M_{ini}	2.413	0.6080	3.580	0.4127	2.701	0.6702	4.154	0.4527
$M_{dilated}$	2.483	0.6092	3.712	0.4222	2.676	0.6688	4.108	0.4507
M_{pred} w/o Mask Dilation Module	2.484	**0.6097**	3.705	0.4231	**2.755**	**0.6782**	**4.231**	**0.4589**
M_{pred} w/ Mask Dilation Module	**2.686**	0.6093	**3.715**	**0.4505**	2.669	0.6677	4.127	0.4505

The bold values indicate the highest performance scores among the methods compared in each row.

method M_{pred}. Additionally, (e) and (f) depict the inpainting result and its zoomed-in view using M_{ini}, (g) and (h) show the inpainting result and its zoomed-in view using uniformly dilated M_{ini}, and (i) and (j) present the inpainting result and its zoomed-in view using M_{pred}. The areas highlighted with red circles are particularly notable, where our method generated superior restoration results compared to the baseline approaches. In the case of M_{ini} and uniformly dilated M_{ini}, they often failed to fully cover the regions requiring restoration and may have included unnecessary areas. This demonstrates the effectiveness of masked region optimization using our proposed method for inpainting.

B. THE EFFECTS OF THE MASK DILATION MODULE

To investigate the effects of the Mask Dilation Module on inpainting quality, Table 2 shows the performance of inpainting with various masks. The table provides results for four types of masks: Masks created by uniformly dilating M_{ini}; Masks generated by the rule-based dilation module described in Section III-D ($M_{dilated}$); Masks predicted by a model trained with uniformly dilated M_{ini} as training data (M_{pred} w/o Mask Dilation Module); and Masks predicted by a model trained with $M_{dilated}$ as training data (M_{pred} w/ Mask Dilation Module). In this experiment, the EdgeConnect method was adopted as the inpainting technique during training. The experimental results showed that for the "Animal Dataset," higher accuracy was achieved with the Mask Dilation Module, except for TOPIQ. In contrast, for the "Object Dataset," better accuracy was obtained without the module.

In the Animal Dataset, where object boundaries (e.g., the fur of cats) were complex and irregular, dilation led to simplified, smoother masks that helped improve restoration. In contrast, objects in the Object Dataset often had simpler, well-defined contours. In such cases, excessive dilation could undesirably cover surrounding contextual pixels, thereby degrading the inpainting quality. This finding aligned with previous studies [35], [36] reporting that unnecessarily large or irregular masks can harm inpainting performance due to loss of structural guidance. Thus, we positioned the Mask Dilation Module not as a required component for all settings but as a selective mechanism for enhancing mask quality in cases involving complex boundary conditions or low-quality initial masks. Designing an algorithm that suppresses excessive mask expansion or contracts the mask when the initial mask is overly large may enable this module to evolve beyond its role as an auxiliary component, thereby contributing to more robust and adaptive mask refinement. Despite these differences, as shown in Table 1, our proposed method outperformed the baseline approaches across both datasets, indicating that our method is not dependent on specific contexts.

TABLE 3. Comparison Based on Inpainting Loss

Loss	Q-Align (↑)	TOPIQ (↑)	LIQE (↑)	MANIQA (↑)
Maximize $S_{inpainted}$	2.746	0.6775	4.213	0.4563
L_I	**2.755**	**0.6782**	**4.231**	**0.4589**

The bold values indicate the highest performance scores among the methods compared in each row.

C. COMPARISON BASED ON INPAINTING LOSS

In the proposed method, when calculating the inpainting quality loss $\mathcal{L}_I$, the loss is defined as the difference between the quality assessment scores of the input image I_{input} and the inpainted image $I_{inpainted}$, denoted as S_{input} and $S_{inpainted}$, respectively. However, a more straightforward approach would be to directly optimize $S_{inpainted}$ by maximizing it during training. Therefore, to investigate the impact of different $\mathcal{L}_I$ calculation methods on the results, we trained models using each approach. The results are shown in Table 3. This experiment was conducted using the "Object Dataset" without utilizing the Mask Dilation Module, and the EdgeConnect method was employed as the inpainting technique during training. As can be seen in Table 3, the approach using the difference between scores achieved higher accuracy, indicating that the proposed loss function leads to improved inpainting quality.

D. USER STUDY

To evaluate the effectiveness of repair region optimization on the inpainting results, we conducted a user study. The participants were informed that the results of this experiment may be published in a paper, and their consent was obtained. We performed inpainting on 20 images using both Initial Masks and masks generated by our proposed method in a pairwise manner. Subsequently, 11 participants were asked to rate the naturalness of the resulting images on a scale from 1 to 10, where a score closer to 1 indicated unnaturalness and a higher score represented greater naturalness. The average scores for each image were aggregated, and the results were visualized using a box plot, as shown in Fig. 5. As shown in Fig. 5, the average score of the Initial Masks was approximately 4.8, which was significantly lower compared to our proposed method's average score of approximately 6.7. Furthermore, in the results of our proposed method, the minimum score was 4.0, indicating that all scores maintained a consistent level of quality. In contrast, the Initial Masks had a minimum score of 2.09, with some results being evaluated as unnatural by the participants. Therefore, it was observed that the scores for the results generated with the masks from our proposed method were consistently higher compared to those generated with the Initial

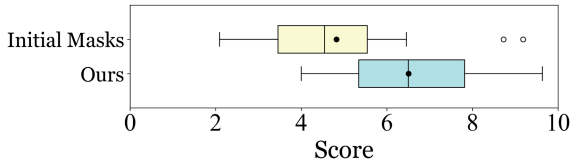

FIGURE 5. Box plot illustrating the results of the user study, comparing the score distributions between the initial masks and our method. The black dot indicates the mean, and open circles represent outliers.

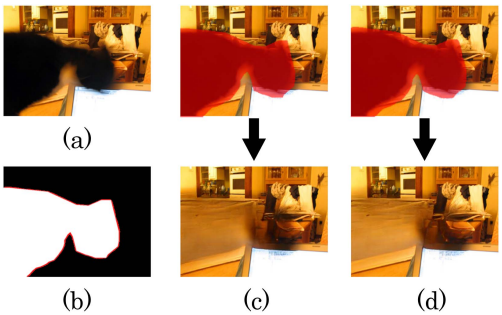

FIGURE 6. Examples of Inpainting Failures: (a) Original Image I_{input}, (b) the difference between the initial mask M_{ini} and the generated mask M_{pred}, (c) Inpainting Results Using M_{ini}, and (d) Inpainting Results Using the M_{pred}.

Masks. These findings demonstrated that using our proposed method can lead to inpainting results that appear more natural.

E. LIMITATIONS

In the previous section, we validated the effectiveness of the proposed method through experiments. However, one of the challenges of the proposed method is that it may fail to restore an image, as illustrated in Fig. 6. In this Fig. (a) shows the original image I_{input}, (b) displays the difference between the initial mask M_{ini} and the generated mask M_{pred}, (c) presents the inpainting result using M_{ini}, and (d) shows the inpainting result using M_{pred}. As shown in the figure, the inpainting results using M_{pred} appear unnatural. In (b), the differences between M_{ini} and M_{pred} are highlighted in red, but there are almost no differences, indicating that the optimization of the masked region has failed. Such failures are more likely to occur when the area to be restored is large and the boundaries of the target area are unclear. In particular, as shown in Fig. 6, when objects in the region to be restored are blurred, the boundaries become unclear, which makes failure more likely. This was likely due to the improper generation of training data during the learning process. Optimizing the masked region appropriately in such cases remains a challenge for future work.

Another notable limitation is that the proposed method may not handle disconnected object regions well when they are occluded by large structures. For example, as shown in the third row of Fig. 4, a red car inside a truck is only partially removed because the portion visible through the front window remains. Since the current dilation approach is based on local information around the masked region, it is not designed to include semantically connected but spatially disconnected areas. This may limit the ability of the system to handle such complex occlusion cases.

VI. CONCLUSION

In this paper, we propose a method for deforming the masked region of an image to improve the performance of image inpainting. Specifically, we present an optimization method for the masked region in deep-learning-based image inpainting using NR-IQA. Furthermore, we compared our proposed method with baseline methods and confirmed qualitative and quantitative improvements in accuracy. In addition, the results of the user study demonstrated that the proposed method significantly enhanced restoration quality. Future work should address large inpainting areas with ambiguous boundaries, which can cause artifacts or unnatural appearances. Enhancing the Mask Dilation Module for better handling of mask dilation and erosion is also essential. Additionally, a future research direction could involve exploring context-aware mask prediction for semantically related but spatially disconnected regions.

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